

A Scoping Review on Immersive Technologies in the Oil and Gas Industry

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ABSTRACT

This paper explores the role of immersive technologies, such as Augmented Reality, Virtual Reality, Mixed Reality, Digital Twins, and Building Information Modeling in the oil and gas industry. Through a comprehensive scoping review, we address key research questions related to the types of immersive technologies used in the context of the oil and gas pipeline industry, and the application scenarios like training, testing, maintenance, and monitoring. It highlights the potential of immersive technologies, identifies research gaps, and suggests future directions.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality; General and reference—Document types—Surveys and overviews

1 INTRODUCTION

In the realm of the oil and gas (O&G) industry, where challenges in construction, safety monitoring, and resource management abound, there is a growing curiosity about the potential of immersive technologies. These include Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR), and digital innovations like Digital Twins (DT) and Building Information Modeling (BIM).

This paper delves into the current research trends of VR, AR, MR, DT, and BIM within the O&G pipeline industry. Our exploration is based on a comprehensive literature review, during which we identify key application scenarios and address the potential benefits of these technologies. Within the scope of this paper, we intend to focus on the following research questions (RQs):

- **RQ1:** What types of immersive technologies have been utilized or investigated in the O&G pipeline context?
- **RQ2:** What are the application scenarios within the O&G industry where immersive technologies have been implemented?

2 METHODOLOGY

The methodology adopted in our paper adheres to the PRISMA systematic review method, analyzing the impact of immersive technologies in the O&G industry. Initiating with an exploratory search, we identified 53 relevant papers, establishing a focused keyword

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set for these technologies in the O&G pipeline industry—“Pipeline” AND (“Oil” OR “Gas”) AND (“Virtual Reality” OR “Augmented Reality” OR “Mixed Reality” OR “Digital Twin”). This search spanned seven key digital libraries, e.g., IEEE and ACM, targeting publications from 2013 to 2023, and yielded 2,045 papers. Using a thorough selection process focusing on title and abstract relevance, we narrowed down to 45 papers, including conference papers and journal articles, for detailed analysis. This streamlined approach enabled a focused examination of the use of immersive technologies (*Technology*) and the application areas (*Scenario*) within the O&G sector, which are listed below.

- **Scenario:** Four subcategories to analyze the specific contexts in which immersive technologies were utilized.
 - **Testing:** Scenarios involved in the systematic process of evaluating the performance of equipment to assess its functionality, identify potential issues, and optimize its operation.
 - **Monitoring:** Scenarios designed to improve real-time data visualization, situational awareness, virtual commissioning, design reviews, and decision-making broaden the scope of monitoring with the addition of strategic planning to optimize industrial operations and maintenance.
 - **Maintenance:** Scenarios designed for equipment inspections and repairs, utilizing immersive technologies to facilitate predictive maintenance, asset integrity checks, and risk management.
 - **Training:** Scenarios leveraged immersive technologies for workforce training, including safety procedures, equipment operation, and emergency simulations.
- **Technology:** The identified technologies, including VR, AR, MR, DTs, and BIM, were chosen for their relevance and specific applications in the O&G pipeline industry.
 - **VR:** Immersive computer-generated environments, transporting users to simulated scenarios pertinent to the O&G industry, and facilitating training, testing, maintenance, and monitoring.
 - **AR:** Digital information overlaid onto the real world, enhancing the perception of the O&G environment with real-time data for improved guidance, visualization, and information delivery.
 - **MR:** Merging virtual and real worlds, including various extended reality (XR) technologies, combines the immersive aspects of VR with real-time data overlay of AR.
 - **DT:** Virtual replicas mirroring physical O&G systems, enabling real-time monitoring, predictive analysis, and simulations for enhanced decision-making and maintenance strategies.
 - **BIM:** Digital representations of physical structures, primarily aiding in planning, designing, and managing pipelines, plants, and related infrastructures in the sector.

3 RESULTS

Our review analyzed the use of five immersive technologies in the O&G industry across 45 papers, focusing on four distinct application scenarios.

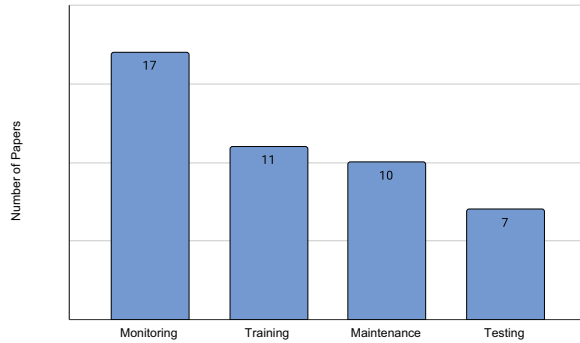


Figure 1: The distribution of different scenarios in O&G.

Scenario This study categorizes the use of immersive technologies in the O&G pipeline industry into four main scenarios (Figure 1). Monitoring (17 papers) was the most common application, highlighting immersive systems' role in improving situational awareness and decision-making through enhanced visualization. Training (11 papers) and maintenance (10 papers) scenarios show the importance of VR, AR, MR, DT, and BIM technologies in providing realistic simulations and supporting predictive maintenance. Testing (7 papers) focused on the evaluation and streamlining of O&G operations, revealing these technologies' effectiveness in streamlining operational performance.

In the Monitoring area, VR and DTs enhanced data visualization and inspection, with studies by Liu et al. [7] and Barbosa et al. [1] on point cloud analysis and 3D underwater inspections. Zhao et al. [12] highlighted DTs in real-time hydropower plant simulation. The Training papers showed VR's role in creating realistic environments for skill training. Sulaiman et al. [11] and Liu et al. [6] demonstrated the use of VR in offshore safety and emergency training, while Chin et al. [4] developed VR simulators for ROV pilot training, highlighting VR's effectiveness in improving safety and proficiency. The Maintenance domain benefits from immersive technology in asset management and integrity checks. Ma et al. [8] and Martinez et al. [9] demonstrate VR's role in subsea data collection and DTs in simulating industrial processes, highlighting their value in predictive maintenance and risk management. In the Testing domain, DTs emerge as a groundbreaking approach for system evaluation and risk analysis. Bu et al. [2] and Priyanka et al. [10] highlight the application of DTs in real-time control valve testing and pipeline project risk assessments. Furthermore, AR technologies, as noted by Chi et al. [3] and Li et al. [5], enhance situational awareness and safety assessment through virtual models and tracking devices. This review highlights how immersive technologies contribute to the O&G industry by improving safety, efficiency, and decision-making. Offering a comprehensive toolkit for tackling operational complexities, from design to decommissioning, it emphasizes the necessity for ongoing research and development in these technologies.

Technology Figure 2 illustrates that DTs are a leading technology in the O&G research, comprising 36% (16) of the papers, and indicates a strong interest in DTs for improving operational efficiency and decision-making. VR and AR each represent 28% (13) of the papers, highlighting their importance in creating immersive experiences for training and data visualization. Meanwhile, the utilization of BIM and MR was relatively limited. This suggests a need for further research into their potential. However, it may also be due to missing keywords related to BIM in our search process.

4 CONCLUSION

This review underscores immersive technologies' significant impact on the O&G industry, enhancing training, maintenance, monitoring,

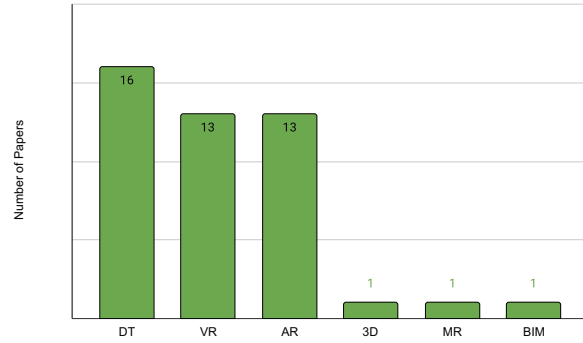


Figure 2: The distribution of immersive technologies in O&G.

and testing for improved safety and efficiency. It addresses their integration into workflows and innovation potential. This sets a direction for further exploration and practical uses in advancing industry standards.

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